COMPLETE STATISTICS

IN ONE VIDEO... PART-II

ConceptsIn5

PART-I ⇒ THEORY

PART-II ⇒ Sums

DATA SCIENCE

TOPICS LIKE COMMENT SUBSCRIBE

* PART-I *

1. Percentile.
2. Box Plot.
3. Central limit theorem.
4. Null & Alternate Hypothesis.
5. Estimates of location.
6. Estimates of variability.
7. Sampling techniques.
8. Confidence interval.
9. Regression To mean.
10. Type I & Type II error.
11. Single & Multiple Regression.
12. Normal Distribution.
13. Poisson Distribution.
14. ANOVA.
15. Correlation.

PART-II

1) Z-Test. (2-Tail, 1-Tail)
2) t-Test.
3) Chi-square Test.
4) Std. deviation, Mean, Std. error, Confidence interval.
5) Linear Regression
6) Correlation...
7) Normal Distribution.
8) Binomial Distribution
9) Poisson Distribution
10) Frequency Distribution Table.....

(*) One-Way ANOVA.
Friedman
Kruskal's

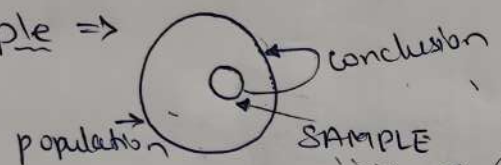
SAMPLING TEST

1)

1) F-Test 2) t-test 3) Chi-square 4) ANOVA

5) Friedman 6) Kruskal's

Sample =>



Example : MUSA → ① Before providing Q.B

② After providing Q.B

* MEAN of SAMPLE's COMPARE.

* For Comparing MEANS of 2 samples

→ F-test, t-test, chi-square

$\mu = 8.5$	$\mu > 8.5$
$\mu \neq 8.5$	$\mu < 8.5$
TWO TAIL	ONE TAIL

Prerequisits :

① Null Hypothesis => (H_0)

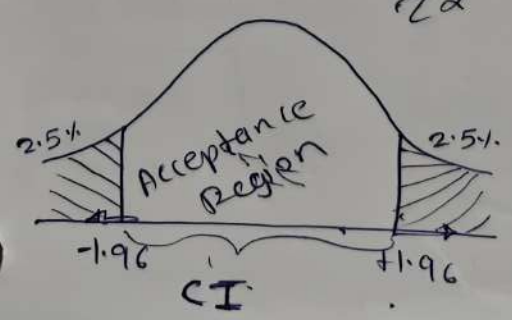
② Alternate Hypothesis => (H_a)

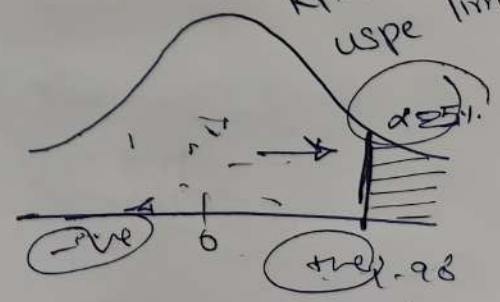
Level of significance => α

eg. $\alpha = 5\%$

eg. $\mu = 100$, $\mu > 100$

Kitna badhega use limit...





① Z-test ①

MAY-23 Q.3 (b)

Q. The average score on a test is 80 with standard deviation of 10. With a new teaching curriculum introduced it is believed that this score will change. On random testing, the score of 38 students, the mean was found to be 88. With a 0.05 significance level, is there any evidence to support this claim?

Given :- ②

$n = 38$ $\mu = 80$ $\sigma = 10$

$\bar{x} = 88$ $\alpha = 0.05$

*** Z-test ***

- $n \geq 30$
- $Z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
- $Z_{cal} < Z_\alpha$

$H_0 \Rightarrow$ Accept ✓

- **STEPS :-**

① N.H (H_0) $\Rightarrow \mu =$.
② A.H (H_a) $\Rightarrow \mu \neq$ ..
③ Test Statistics $\Rightarrow Z_{cal}$
④ Level of Significance $\Rightarrow \alpha$
④ Critical value .. Z_α .
⑤ Decision.

STEP 1 :-

$H_0 \Rightarrow \mu = 80$

$H_a \Rightarrow \mu \neq 80$

STEP 2 :-

$$Z_{cal} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{88 - 80}{10/\sqrt{38}}$$

$$\boxed{Z_{cal} = 4.93}$$

STEP 3 :-

L.O.F $\Rightarrow \alpha = 0.05$

STEP 4 :-

$Z_{0.05} = \pm 1.96$

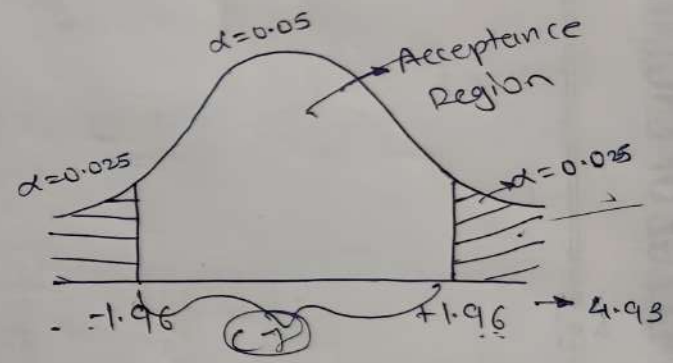
STEP 5 :-

Decision

$Z_{cal} \not< Z_\alpha$

$(4.93) \not< (1.96)$

$\Rightarrow$ H_0 is Rejected...



CLAIM :- That the value will change..

IS $\Rightarrow$ true

② * t-test * ②

ONE-TAIL

Q. A fitness coach believes a new diet plan reduces avg. body fat percentage. The previous mean body fat = 28%. A sample of 18 athletes follows new diet which has : * mean = 25.4% (fat) * ~~σ = 4.6~~ Std. deviation = 4.6

* t-test *

$n < 30$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$s = \sqrt{\frac{\Sigma (x-\bar{x})^2}{n}}$$

$t(\alpha, df)$

$df = n-1$

① H_0, H_a

② t_{cal}

③ α, df

④ $t(\alpha, df)$

⑤ Decision

* GIVEN :- $n = 18$, $\alpha = 5\%$, $s = 4.6$, $\mu = 28$, $\bar{x} = 25.4$

① $H_0 \Rightarrow \mu = 28$

$H_a \Rightarrow \mu < 28$

Reduce fat

② $$t_{cal} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{25.4 - 28}{4.6/\sqrt{18}}$$

$\boxed{\cancel{t_{cal} = -1.740}}$ $\boxed{t_{cal} = -2.318}$

③ $\alpha = 0.05$ $df = n-1 = 18-1$

$df = 17$

$t(0.05, 17)$

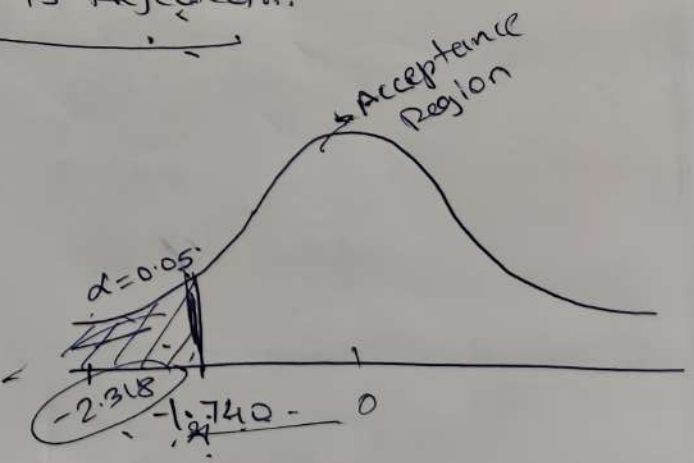
④ $t(\alpha, df) = -1.740$

⑤ Decision :-

$|t_{cal}| > |t_{critical}|$

(-2.318) (-1.740)

$\therefore H_0$ is Rejected...



May 25 Q.2 (b). ③ * Chi-Square test * ③ Non-parametric

Q. What is chi-square test. Based on gender whether they prefer Online or Instore shopping? n = 200 customers. $\alpha = 5\%$

Gender	Prefer Online Shopping	Prefers Instore Shopping	Total
Male	60	40	100
Female	70	30	100
Total	130	70	200

$$\chi^2_{cal} = \frac{(O-E)^2}{E}$$

$$E = \frac{RT \times CT}{T}$$

α

$df = (col-1)\cdot(rows-1)$

② $\alpha = 0.05$ (0.05, 1)

$df = (col_s - 1)(rows - 1)$

$df = (2-1)(2-1) = 1$

③ $\chi^2(0.05, 1) = 3.841$

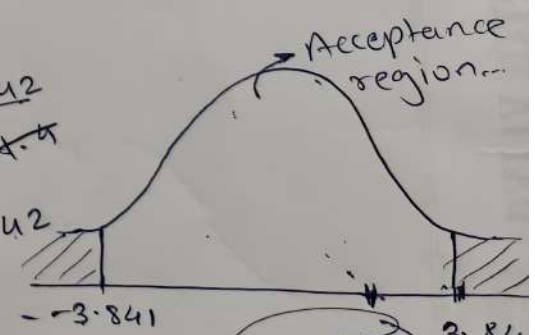
④ Decision :-

$\chi^2_{cal} < \chi^2_{critical}$

→ Ho is Accepted...

Online / Instore	O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
M/On	60	65	25	0.3846
M/In	40	35	25	0.7142
F/On	70	65	25	0.3846
F/In	30	35	25	0.7142

χ^2_{cal} ~~3.569~~ 2.1976



~~3.569~~ 2.1976

* MEAN, Standard deviation, Standard Error & Confidence Interval *

	x	$x-\bar{x}$ ($\bar{x}=3.8$)	$(x-3.8)^2$
1	4	0.2	0.04
2	2	−0.8	0.64
3	2	−1.8	3.24
4	3	1.2	1.44
5	5	−1.8	3.24
6	2	1.2	1.44
7	6	−1.8	3.24
8	2	−1.8	3.24
9	7	3.2	10.24
10	6	2.2	4.84
	38		Σ 31.6

① **Mean** $= \frac{\Sigma x}{n}$

$$= \frac{4+2+2+3+5+2+6+2+7+6}{10}$$

$$\boxed{\text{Mean} = 3.8}$$

② **Standard Deviation:**

$$SD = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{31.6}{9}}$$

$$\boxed{SD = 1.874}$$

③ **Standard error:-**

$$SD.\ error = \frac{S.D}{\sqrt{n}}$$

$$SD\ error = \frac{1.874}{\sqrt{10}}$$

$$\boxed{SD.\ error = 0.592}$$

④ **Confidence Interval:-**

$$CI = \bar{x} \pm t_{\alpha/2} \times \frac{S}{\sqrt{n}}$$

$$CI = \bar{x} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

for $t \Rightarrow \alpha = \alpha/2$

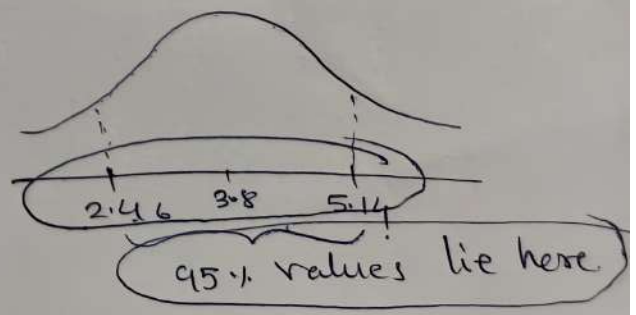
$df = n-1$

$n < 30$

$$CI = 3.8 \pm t(0.025, 9) \times \cancel{0.592}\ \frac{1.874}{\sqrt{n}}$$

$$= 3.8 \pm 1.339$$

$$\boxed{CI = (2.46,\ 5.14)}$$



May-24 Q.5 (b)

* LINEAR REGRESSION of LINE *

WITH_CORRETATION _COEfficent ....

$y = B_0 + B_1 \cdot x$

	X	Y	XY	X^2	Y^2
1	2	69	138	4	4761
2	9	98	882	81	9604
3	5	82	410	25	6724
4	5	77	385	25	5929
5	3	71	213	9	5041
6	7	84	588	49	7056
Σ	31	481	2616	193	39115

$$y = B_0 + B_1 \cdot x$$

$$B_1 = \frac{n \Sigma xy - (\Sigma x \cdot \Sigma y)}{n \Sigma x^2 - (\Sigma x)^2}$$

$$B_0 = \bar{y} - B_1 \cdot \bar{x}$$

$$= \frac{\Sigma y}{n} - B_1 \cdot \frac{\Sigma x}{n}$$

$$r = \frac{\Sigma xy}{\sqrt{(\Sigma x^2) \cdot (\Sigma y^2)}}$$

$$B_1 = \frac{n \Sigma xy - (\Sigma x \cdot \Sigma y)}{n \Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{(6 \times 2616) - (31 \times 481)}{(6 \times 193) - (31)^2}$$

$$\boxed{B_1 = 3.99}$$

$$B_0 = \bar{y} - B_1 \cdot \bar{x}$$

$$= \frac{\Sigma y}{n} - B_1 \cdot \frac{\Sigma x}{n}$$

$$= \frac{481}{6} - \left(3.99 \times \frac{31}{6}\right)$$

$$\boxed{B_0 = 59.55}$$

$$y = 59.55 + 3.99x$$

14th Week ki sale, etc.

$x = 14$

$y = 59.55 + (3.99 \times 14)$

$$\boxed{y_{14} = 115.4116}$$

$$r = \frac{\Sigma xy}{\sqrt{(\Sigma x^2)(\Sigma y^2)}}$$

$$= \frac{2616}{\sqrt{193 \times 39115}}$$

$$\boxed{r = 0.9521}$$

* PROBABILITY *

* $m = np$ = mean
* n = no. (Total)
* $x \rightarrow$ mean (avg)
* $\mu \rightarrow$ claim
* $\sigma \rightarrow$ std. deviation.

① Normal Distribution ①

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)}$$

Q. 4 (a) MAY - 23

Speed of car normally distributed with mean 90 km/hr. $\sigma = 10$ km/hr. What is probability of travelling more than 100 km/hr.

$\mu = 100$ $\quad x = 90$

$\mu > 100$ $\quad \sigma = 10$

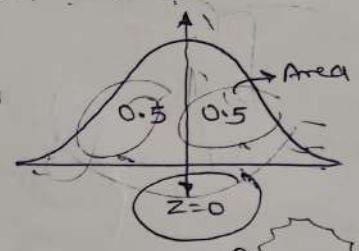
$$f(x) = \frac{1}{10\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{90-100}{10}\right)}$$

$$= \frac{e^{\frac{1}{2}}}{10\sqrt{2\pi}}$$

$$\boxed{f(x) = 0.0 6579}$$

* Normal Distribution *

$$Z = \frac{x - \mu}{S.D}$$ → mean



* Area based probability...

Q.4 a) May-23

Speed of car normally distributed with mean 90 km/hr. S.D = 10 km/hr. What is probability of travelling more than 100 km/hr.

⇒ $\mu = 90$ km/hr, $P(x \geq 100) = ?$ $\sigma = 10$

$$Z = \frac{x - \mu}{\sigma} = \frac{100 - 90}{10} = 1$$

$\boxed{Z = 1}$

$P(Z \geq 1) = ?$ ⇒ 15.8% probability.

We have, $P(Z \leq 1) = 0.8413$

$\therefore P(Z > 1) = 1 - P(Z \leq 1)$

$= 1 - 0.8413$

$\boxed{P(Z \geq 1) = 0.158}$

* Binomial Distribution *

$$P(X = x) = {}^nC_x \cdot p^x \cdot q^{n-x}$$

$p + q = 1$. mean = np variance = npq

Q. Observed 40% of the students wear certain glasses. At random, what is chance: $n = 5$

① No one wear glasses: $x = 0$

$p = 0.40$, $q = 0.6$, $n = 5$ ⇒ Given

$P(x=0) = {}^5C_0\, p^0\, q^{5-0}$

$= 1 \cdot 1 \cdot (0.6)^5$ → $\frac{5!}{5! \times 0!}$, $0! = 1$, $p^0 = 1$

$\boxed{P(x=0) = 0.0776}$ 7.76%

② Atleast one wear glasses:

$P(x \geq 1) = 1 - P(x < 1)$

$= 1 - P(x = 0)$

$= 1 - 0.0776$

$\boxed{P(x \geq 1) = 0.9224}$ 92.24%

* Poisson Distribution *

$$P(X=x) = \frac{e^{-m} m^{x}}{x!}$$

mean = variance = $\boxed{np = m}$

Q. No. of accidents of truck drivers follows poisson distribution. with mean = 3. Out of 1000 taxi drivers find approx no. of drivers with:

① No accident in a year.

② Three or more accidents in a year.

- Given :- $e^{-1} = 0.3679$, $e^{-2} = 0.1353$, $e^{-3} = 0.0498$, $m = 3$

$$P(X=x) = \frac{e^{-m} m^{x}}{x!}$$

$0! = 1$

① $P(X=0) = \frac{e^{-3} \cdot 3^{0}}{0!} = 0.0498$

② $P(X \geq 3) = 1 - [P(X<3)]$

$= 1 - [P(X=0) + P(X=1) + P(X=2)]$

$= 1 - [0.0498 + \frac{e^{-3} \cdot 3^{1}}{1!} + \frac{e^{-3} \cdot 3^{2}}{2!}]$

$P(X \geq 3) = \cancel{0.5767}\ \boxed{0.3486}$

* To find Number of Accidents of drivers :-

① No accident in a year :-

= Total no. × prob

= 1000 × 0.0498

= 49.8

≈ 50 drivers

② 3 or more :

= Total no. × prob.

= 1000 × ~~0.5767~~

= ~~57.67~~ 0.3486

~~≈ 586 drivers~~

~~≈ 577 drivers~~

≈ 347 drivers

* STEM & LEAF *

124 | 12 | 4
58 | 6
586

Age :- 20, 23, 27, 32, 34, 35, 36, 37, 41, 41, 44, 47, 47, 47, 49, 53, 53, 55, 58, 59, 98

∵ values are already arranged....

Stem	Leaf
2	0, 3, 7
3	2, 4, 5, 6, 7
4	1, 1, 4, 7, 7, 7, 9
5	3, 3, 5, 8, 9
9	8

Outlier...

ANOVA

—xox—xox—

ANALYSIS OF VARIANCE

- also called F-test, F-Statistics
- used for comparing MEAN of more than 2 samples.

Types :-

*** 1-WAY ANOVA ***

- One Response is affected by one factor.

Factor → Response

class 1 → avg. score

class 2 → avg. score

*** 2-WAY ANOVA ***

- One Response is affected by 2 factors

Factor 1 → Response variable

Factor 2 → Response variable

—xox—

1-WAY ANOVA

A	B	C
25	31	24
30	39	30
36	38	28
38	42	25
31	35	30

STEPS

1) STEP 1 :

Null Hypothesis : $\mu_1 = \mu_2 = \mu_3$ (H_0)

Alternate Hypo : $\mu_1 \neq \mu_2 \neq \mu_3$ (H_a)

2) STEP 2 :

State α {if not given : 0.05}

$a = 3$ $N = 15$ $n = 5$

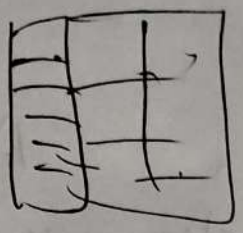
3) STEP 3 :

Degree of freedom : $df_B = a - 1$ Between

$df_W = N - a$ Within

$df_T = N - 1$ Total

$25^2 + 30^2 +$



$a \rightarrow$ no. of columns. (chota)

$n \rightarrow$ no. of rows. (medium)

$N \rightarrow$ Total no. of elements. (Bigger)

$T \rightarrow$ Addition of N values

$\Sigma y^2 \rightarrow$ Summation of Square of each term.

$F_{critical} = F(df_B, df_W)$

4) STEP 4 :

Test Statistics

$$F_{cal} = \frac{MSB}{MSW}$$

F_{cal}

- $MSB = \frac{SS_B}{df_B}$
- $MSW = \frac{SS_W}{df_W}$
- $SS_B = \frac{\Sigma(\Sigma a_i)^2}{n} - \frac{T_i^2}{N}$
- $SS_W = \Sigma y^2 - \frac{\Sigma(\Sigma a_i)^2}{n}$
- $SS_T = \Sigma y^2 - \frac{T_i^2}{N}$

5) STEP 5 :

Decision

if $F_{cal} < F_{critical}$

then Null Hypothesis is Accepted...

सा विद्या या विमुक्तये

Date :

1-WAY ANOVA

No.	A	B	C
1	25	31	24
2	30	39	30
3	36	38	28
4	38	42	45
5	31	35	28

*** STEP 1 :-**

Null Hypothesis (H_0) $\Rightarrow \mu_1 = \mu_2 = \mu_3$

Alternate Hypo (H_a) $\Rightarrow \mu_1 \neq \mu_2 \neq \mu_3$

*** STEP 2 :-**

$\alpha = 0.05$

*** STEP 3 :-**

$df_B = a - 1 = 3 - 1 = 2$

$df_W = N - a = 15 - 3 = 12$

$df_T = N - 1 = 15 - 1 = 14$

*** STEP 4 :-**

$F_{critical} \Rightarrow F(2, 12)$

$\boxed{F_{critical} = 3.89}$

*** STEP 5 :-**

$$F_{cal} = \frac{MSB}{MSW}$$

$$MSB = \frac{SSB}{df_B}$$

$$\therefore SSB = \frac{\sum (\sum a_i)^2}{n} - \frac{T_i^2}{N}$$

$a_1 = 25 + 30 + 36 + 38 + 31 = 160$

$a_2 = 31 + 39 + 38 + 42 + 35 = 185$

$a_3 = 24 + 30 + 28 + 45 + 28 = 135$

$T_i = a_1 + a_2 + a_3 = 160 + 185 + 135 = 480$

दिल्ली कम्प्यूटर यांत्रिकी महाविद्यालय DMCE सा विद्या या विमुक्तये

$$SS_B = \frac{\Sigma(\Sigma a_i)^2}{n} - \frac{T^2}{N}$$

$$= \frac{160^2 + 185^2 + 135^2}{5} - \frac{480^2}{15}$$

$$= 15610 - 15360$$

$$\boxed{SS_B = 250}$$

$$MS_B = \frac{SS_B}{df_B} = \frac{250}{2}$$

$$\boxed{MS_B = 125}, \text{ ---- } (1)$$

$$SS_W = y^2 - \frac{\Sigma(\Sigma a_i)^2}{n}$$

$$y^2 = 25^2 + 30^2 + 36^2 + 38^2 + 31^2 + 31^2 + 39^2 + 38^2 + 42^2 + 35^2 + 24^2 + 30^2 + 28^2 + 25^2 + 28^2$$

$$\boxed{y^2 = 15810}$$

$$SS_W = 15810 - 15610$$

$$SS_W = 200$$

$$MS_W = \frac{SS_W}{df_W} = \frac{200}{12}$$

$$\boxed{MS_W = 16.667}, \text{ --- } (2)$$

THANK YOU

$$F_{cal} = \frac{MS_B}{MS_W} = \frac{125}{16.667} = 7.5$$

* STEP 6 :- DECISION

F_{cal} ≯ $F_{critical}$

(7.5) ≯ (3.89)

∴ Null Hypothesis is Rejected.